

BMML – Theory 1

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Problem 1: Simplify the following expressions:

1.1 (a)

$$\begin{aligned}\|uv^T - A\|_F^2 - \|A\|_F^2 &= \text{tr}\left((uv^T - A)^T(uv^T - A)\right) - \text{tr}(A^T A) = \\ &= \text{tr}\left((uv^T)^T(uv^T)\right) - 2\text{tr}(A^T(uv^T)) + \text{tr}(A^T A) - \text{tr}(A^T A) \\ &= \text{tr}(vu^T uv^T) - 2\text{tr}(A^T uv^T) = \\ &= \text{tr}(v(u^T u)v^T) - 2u^T A v = \\ &= \|u\|^2 \|v\|^2 - 2u^T A v\end{aligned}$$

1.2 (b)

$$\begin{aligned}&\text{tr}((2I_n + aa^T)^{-1}(uv^T + vu^T)) = \\ &= \text{tr}\left(\left[\left(\frac{1}{2}I_n - \left(\frac{1}{4}\right)a\left(1 + \frac{1}{2}a^T a\right)^{-1}1a^T\right)(uv^T + vu^T)\right]\right) \\ &= \text{tr}\left(\left[\left(\frac{1}{2}I_n - \frac{aa^T}{4 + 2a^T a}\right)(uv^T + vu^T)\right]\right) \\ &= \text{tr}\left(\frac{1}{2}uv^T + \frac{1}{2}vu^T - \frac{aa^T uv^T}{4 + 2a^T a} - \frac{aa^T vu^T}{4 + 2a^T a}\right) \\ &= \frac{1}{2}\text{tr}(uv^T) + \frac{1}{2}\text{tr}(vu^T) - \frac{1}{4 + 2a^T a}[\text{tr}(aa^T uv^T) + \text{tr}(aa^T vu^T)] \\ &= \frac{1}{2}v^T u + \frac{1}{2}u^T v - \frac{1}{4 + 2a^T a}[(v^T a)(a^T u) + (u^T a)(a^T v)] \\ &= u^T v - \frac{2(a^T u)(a^T v)}{4 + 2a^T a} \\ &= u^T v - \frac{(a^T u)(a^T v)}{2 + a^T a}\end{aligned}$$

1.3 (c)

$$\sum_{i=1}^n \langle S^{-1}a_i, a_i \rangle = \sum_{i=1}^n a_i^T S^{-1}a_i = \text{tr}\left(S^{-1} \sum_{i=1}^n a_i a_i^T\right) = \text{tr}(S^{-1}S) = d$$

Problem 2: For each of the following functions, find its first derivative using the technique of differentials:

2.1 (a)

Let $M(t) = A - tI_n$, so that: $f(t) = \det(M(t))$ and $M'(t) = -I_n$.

And we know that:

$$\begin{aligned} \frac{d}{dt} \det(M(t)) &= \det(M(t)) \cdot \operatorname{tr}(M(t)^{-1} M'(t)) \Rightarrow \\ \Rightarrow \frac{d}{dt} \det(A - tI_n) &= -\det(A - tI_n) \cdot \operatorname{tr}((A - tI_n)^{-1}) \end{aligned}$$

2.2 (b)

Let $g(t) = (A + tI_n)^{-1}b$;

$$\begin{aligned} \frac{dg(t)}{dt} &= -(A + tI_n)^{-1}(A + tI_n)^{-1}b \\ f(t) &= \|g(t)\| = (g(t)^T g(t))^{\frac{1}{2}} \\ \frac{df}{dt} &= \frac{g(t)^T \frac{dg(t)}{dt}}{\|g(t)\|} = \\ &= -\frac{g(t)^T (A + tI_n)^{-1} (A + tI_n)^{-1} b}{\|g(t)\|} = \\ &= -\frac{b^T (A + tI_n)^{-1} (A + tI_n)^{-1} b}{\|(A + tI_n)^{-1} b\|} \end{aligned}$$

Problem 3: For each of the following functions, find its gradient ∇f ;

3.1 (a)

$$\text{Let } A = (2I_n + aa^T)^{-1} \quad \text{and} \quad X = uv^T + vu^T.$$

We want to find $\text{tr}(AX)$.

$$A = (2I_n)^{-1} - (2I_n)^{-1}a(1 + a^T(2I_n)^{-1}a)^{-1}a^T(2I_n)^{-1}$$

$$\Rightarrow A = \frac{1}{2}I_n - \frac{1}{4}a\left(1 + \frac{1}{2}a^Ta\right)^{-1}a^T$$

$$\Rightarrow A = \frac{1}{2}I_n - a\frac{a^T}{4 + 2a^Ta} \quad \text{tr}(AX) = \text{tr}\left(\left(\frac{1}{2}I_n - \frac{aa^T}{4 + 2a^Ta}\right)(uv^T + vu^T)\right)$$

$$\Rightarrow \text{tr}\left(\frac{1}{2}(uv^T + vu^T)\right) - \text{tr}\left(\frac{(aa^T)(uv^T + vu^T)}{4 + 2a^Ta}\right)$$

$$\text{tr}\left(\frac{1}{2}(uv^T + vu^T)\right) = \frac{1}{2}(v^Tu + u^Tv) = u^Tv$$

$$\text{tr}(aa^Tuv^T) = \text{tr}(v^Taa^Tu) = (v^Ta)(a^Tu)$$

$$\text{tr}(aa^Tvu^T) = \text{tr}(u^Taa^Tv) = (u^Ta)(a^Tv)$$

$$\Rightarrow \text{tr}(AX) = u^Tv - \frac{(v^Ta)(a^Tu) + (u^Ta)(a^Tv)}{4 + 2a^Ta}$$

$$\Rightarrow u^Tv - \frac{2(a^Tu)(a^Tv)}{4 + 2a^Ta} \Rightarrow u^Tv - \frac{(a^Tu)(a^Tv)}{2 + a^Ta}$$

3.2 (b)

$$\text{Let } f(x) = \langle x, x \rangle^{<x,x>} = (x^Tx)^{x^Tx}.$$

Define $u(x) = x^Tx = \|x\|^2$. Thus, $f(x) = u^u$.

By the chain rule, the gradient is $\nabla f(x) = \frac{df}{du} \nabla u$.

$$\frac{df}{du} = \frac{d}{du}(u^u) = \frac{d}{du}e^{u \ln u} = e^{u \ln u} * \left(1 * \ln u + u * \frac{1}{u}\right) = u^{u(\ln u + 1)}.$$

$$\nabla u = \nabla(x^Tx) = 2x. \nabla f(x) = [u^{u(\ln u + 1)}] * (2x)$$

$$\Rightarrow \nabla f(x) = \left[(\|x\|^2)^{\|x\|^2}(\ln(\|x\|^2) + 1)\right] * 2x \nabla f(x) = 2(\|x\|^2)^{\|x\|^2}(\ln(\|x\|^2) + 1)x.$$

3.3 (c)

$$f(x) = \|Ax - b\|^p = \left(\sum_{i=1}^m |(Ax - b)_i|^2\right)^{\frac{p}{2}}$$

$$\text{Let } z = Ax - b \Rightarrow f(x) = \|z\|^p \nabla f(x) = p\|z\|^{p-2} \nabla \|z\|^2$$

$$\nabla \|z\|^2 = \nabla(z^Tz) = 2z^T \nabla z$$

$$\nabla z = A \Rightarrow \nabla \|z\|^2 = 2(Ax - b)^T A$$

$$\nabla f(x) = p\|Ax - b\|^{p-2} A^T(Ax - b)$$

Problem 4: For each of the following functions, find all stationary points (i.e. points where the gradient is zero) and indicate conditions when they exist:

4.1 (a)

$$\begin{aligned}
 \nabla(\langle c, x \rangle) &= c \\
 \nabla\left(\frac{\sigma}{3}\|x\|^3\right) &= \sigma\|x\|x \\
 \Rightarrow \nabla f(x) &= c + \sigma\|x\|x \\
 c + \sigma\|x\|x &= 0 \\
 \sigma\|x\|x &= -c \\
 \sigma\|x\|^2 &= \|c\| \\
 \|x\| &= \sqrt{\frac{\|c\|}{\sigma}} \\
 \dot{x} &= -\frac{c}{\sqrt{\sigma\|c\|}} \text{ (always } \exists \text{)}.
 \end{aligned}$$

4.2 (b)

To find the stationary point, we set the gradient of $f(x)$ to zero:

$$\nabla f(x) = a + \frac{b}{1 - \langle b, x \rangle} = 0 \Rightarrow \frac{b}{1 - \langle b, x \rangle} = -a.$$

Taking the dot product with b on both sides yields:

$$\langle b, \frac{b}{1 - \langle b, x \rangle} \rangle = -\langle b, a \rangle \Rightarrow \|b\| \frac{\|b\|^2}{1 - \langle b, x \rangle} = -\langle b, a \rangle.$$

For the logarithm in $f(x)$ to be defined, we need $1 - \langle b, x \rangle > 0$. The equation above shows this is equivalent to $-\frac{\|b\|^2}{\langle b, a \rangle} > 0$. Since $\|b\|^2 \geq 0$, this condition holds if and only if $\langle b, a \rangle < 0$.

Under this condition, we can solve for $\langle b, x \rangle$:

$$\langle b, x \rangle = 1 + \frac{\|b\|^2}{\langle b, a \rangle}.$$

4.3 (c)

The function is $f(x) = \langle c, x \rangle \exp(-\langle Ax, x \rangle)$. To find stationary points, we set its gradient to zero.

Using the product rule, the gradient is:

$$\nabla f(x) = \exp(-\langle Ax, x \rangle)(c - 2\langle c, x \rangle Ax)$$

Since $\exp(-\langle Ax, x \rangle) > 0$, we set the other term to zero:

$$c - 2\langle c, x \rangle Ax = 0 \Rightarrow c = 2\langle c, x \rangle Ax$$

Let the scalar be $\lambda = 2\langle c, x \rangle$. The equation becomes $c = \lambda Ax$. Since A is in $S_{\{++\}}^n$, it's invertible, so $x = (\frac{1}{\lambda})A^{-1}c$. Note $\lambda \neq 0$ because $c \neq 0$.

Substitute x back into the definition of λ :

$$\lambda = 2 \langle c, \left(\frac{1}{\lambda}\right) A^{-1}c \rangle = \left(\frac{2}{\lambda}\right) \langle c, A^{-1}c \rangle$$

$$\Rightarrow \lambda^2 = 2 \langle c, A^{-1}c \rangle$$

$$\Rightarrow \lambda = \pm \sqrt{2 \langle c, A^{-1}c \rangle}$$

This gives two stationary points:

$$x = \pm \frac{A^{-1}c}{\sqrt{2 \langle c, A^{-1}c \rangle}}$$

Existence: Since A is positive definite and $c \neq 0$, we have $\langle c, A^{-1}c \rangle > 0$. Therefore, the two stationary points **always exist** under the given conditions.

Problem 5: Solve the following optimization problem:

$$\min_{X \in S_{++}^n} \sum_{i=1}^m \langle X^{-1} a_i, a_i \rangle + \ln \det(X) \text{ where } a_1, \dots, a_m \in \mathbb{R}^n.$$

We find the stationary point by setting the gradient to zero:

$$\nabla_X f(X) = X^{-1} - X^{-1} \left(\sum_{i=1}^m a_i a_i^T \right) X^{-1} = 0$$

This simplifies to:

$$X^{-1} = X^{-1} \left(\sum_{i=1}^m a_i a_i^T \right) X^{-1}$$

Since $X \in S_{++}^n$, it is invertible. Pre- and post-multiplying by X isolates the solution:

$$X = \sum_{i=1}^m a_i a_i^T$$

This solution is the minimizer. It exists within the domain S_{++}^n if and only if the resulting matrix X is positive definite, which requires that the set of vectors $\{a_1, \dots, a_m\}$ must span $\mathbb{R}^n$.

Problem 6: 6. Let's consider the Principal Component Analysis (PCA) approach. Suppose we are given a training dataset $\{x_i\}_{i=1}^N$, $x_i \in \mathbb{R}^D$ and we want to reduce it to a dataset of lower dimensionality d using projection to a linear subspace that is determined by matrix $P \in \mathbb{R}^{D \times d}$. Orthogonal projection of a vector to this space can be calculated as $P(P^T P)^{-1} P^T x$. Then to find the best matrix P we may consider the following optimization problem:

$$F(P) = \sum_{i=1}^N \left\| x_i - P(P^T P)^{-1} P^T x_i \right\|^2 = N \operatorname{tr} \left(\left(I - P(P^T P)^{-1} P^T \right)^2 S \right) \rightarrow \min_P$$

where $S = \frac{1}{N} \sum_{i=1}^N x_i x_i^T$ is the sample covariance matrix for the normalized dataset.

6.1 (a) Find gradient $\nabla_P F(P)$, calculated for arbitrary matrix with orthogonal columns, i.e., $P: P^T P = I$ (Note: you should first compute differential $dF(P)$ for arbitrary matrix P and only then use orthogonality property of its columns in obtained expression.)

$$\begin{aligned} F(P) &= N \cdot \operatorname{tr} \left[\left(I - P(P^T P)^{-1} P^T \right)^2 S \right] = N \cdot \operatorname{tr} \left[d \left[\left(I - P(P^T P)^{-1} P^T \right)^2 S \right] \right] \\ dF(P) &= N \cdot \operatorname{tr} \left[2 \left(I - P(P^T P)^{-1} P^T \right) \left(-d \left[P(P^T P)^{-1} P^T \right] \right) S \right] \\ d \left[P(P^T P)^{-1} P^T \right] &= dP(P^T P)^{-1} P^T - P(P^T P)^{-1} d(P^T P)(P^T P)^{-1} P^T + P(P^T P)^{-1} dP^T \\ dF(P) &= -2N \cdot \operatorname{tr} \left[\left(I - P(P^T P)^{-1} P^T \right) \left(dP(P^T P)^{-1} P^T - \right. \right. \\ &\quad \left. \left. - P(P^T P)^{-1} d(P^T P)(P^T P)^{-1} P^T + P(P^T P)^{-1} dP^T \right) S \right] \\ P^T P &= I \Rightarrow (P^T P)^{-1} = I \text{ and } d(P^T P) = 0 \\ dF(P) &= -2N \cdot \operatorname{tr} \left[(I - PP^T)(dPP^T + PdP^T)S \right] = \\ &= -2N \cdot [\operatorname{tr}(dPP^T S(I - PP^T)) + \operatorname{tr}(PdP^T S(I - PP^T))] = \operatorname{tr}(\nabla_P F \cdot dP) \\ \nabla_P F(P) &= -2N[SP - PP^T SP] \end{aligned}$$